Scene Graph Database Agent: Hierarchical Spatio-Temporal Scene Graph Driven Agent for Long Video Understanding

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Abstract

Long-form video understanding requires locating specific evidence—often involving particular entities, their relations, and precise temporal context—across videos that may span hours. Recent agentic approaches build a searchable caption database and let a reasoning model iteratively query it; yet captions are flat text without entity or temporal structure, causing semantic drift, redundant searches, and frequent failure to converge. We propose SGDAgent, which augments the caption database with a hierarchical spatio-temporal scene graph index to convert open-ended search into structured candidate verification. The index comprises three levels—frame-level entity–relation graphs capturing spatial snapshots, segment-level temporal interaction graphs aggregating entity dynamics within sliding windows, and a global event graph encoding cross-segment event chains—all linked by a unified entity registry for cross-scale identity consistency. On this index, a graph-driven reasoning agent follows a propose–verify–answer paradigm: graph queries propose spatiotemporally grounded candidate windows, frame-level visual inspection verifies candidates on raw frames, and an uncertainty-aware control layer governs evidence sufficiency and fallback. Under strictly matched settings on LVBench (same backbone, caption source, and budget), SGDAgent raises end-to-end accuracy from 54.8% to 73.0% and reduces the failed-termination rate from 16.5% to 1.6% over the caption-only baseline. Trajectory analysis confirms that the gains stem from restructuring the search process—shifting from open retrieval to graph-anchored candidate verification—rather than from the graph directly producing answers.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Neural networks*; • **Information systems** → **Question answering**.

Keywords

long-form video question answering, scene graph, agentic reasoning, multimodal retrieval, candidate verification

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1 Introduction

Answering questions about long-form videos requires locating specific evidence—often involving particular entities, their relations, and precise temporal context—within a timeline that may span hours. Recent agentic approaches convert the video into a searchable caption database and let a reasoning LLM iteratively query it [**? ? ?**]; yet *the search space is unstructured*: captions are flat text without entity identities, relation bindings, or temporal anchoring, easily causing semantic drift, redundant searches, and trajectories that fail to converge.

Recent works have begun to address this instability by improving the agent's search strategy—through multi-round frame retrieval [**?**], explicit grounding–verification pipelines [**?**], or multi-granularity toolkits [**?**]—yet they all leave the underlying evidence space essentially unchanged: flat captions or raw frames. A common thread is that these methods improve *how the agent searches* without restructuring *what it searches over*. This raises a natural question: can the evidence space itself be reorganized to yield more stable and verifiable search trajectories?

Scene graphs—structured representations that bind entities, attributes, and pairwise relations—offer a natural candidate for such reorganization, as they provide exactly the entity-level and relational grounding that flat captions lack [**? ?**]. Recent video scene graph generation methods further extend this idea to temporally evolving visual relations across frames [**?**]. However, two gaps prevent their direct application to long-form video QA. First, a single-frame scene graph captures only a *spatial* snapshot and cannot represent temporal interactions or cross-segment entity continuity across an extended timeline. Second, no prior work has built a reasoning agent that uses scene graphs as its *primary retrieval index*—one that queries graph structure to propose candidates and verifies them visually, rather than treating graphs as a post-hoc annotation.

We propose **SGDAgent**, which addresses both gaps. On the representation side, we construct a *hierarchical spatio-temporal scene graph index* that augments the caption database with three levels of structured knowledge: frame-level entity–relation graphs that capture per-keyframe spatial snapshots, segment-level temporal interaction graphs that aggregate entity dynamics within sliding windows, and a global event graph that encodes cross-segment event chains with explicit temporal and semantic edges. A shared video-level entity registry links all three levels to maintain identity consistency across scales. On the reasoning side, we build a graph-driven agent that operates in an observe–reason–act loop over this augmented database. At each reasoning step the agent can invoke graph queries at any level to *propose* a small set of spatiotemporally grounded candidate windows, then call frame-level visual inspection to *verify* candidates on raw frames, with caption search serving as a complementary fallback channel. An uncertainty-aware control layer—comprising query-type routing, confidence gating, and

terminal convergence—governs when accumulated evidence is sufficient to commit an answer. This design converts the agent's core search process from open-ended caption retrieval into structured candidate verification: the graph narrows *where to look*, visual inspection confirms *what is there*, and the control layer ensures the trajectory converges reliably.

Under strictly matched settings on LVBench [**?**] (same backbone, caption source, and budget), SGDAgent raises end-to-end accuracy from 54.8% to 73.0% and reduces the failed-to-terminate (FTT) rate from 16.5% to 1.6% compared with the caption-only baseline. Across four benchmarks, SGDAgent with the lighter `o4-mini` backbone consistently approaches or matches DVD with the substantially stronger `o3`, suggesting that the structured graph index compensates for weaker open-ended search capability. Ablation and trajectory analyses confirm that the improvement stems from restructuring the search process itself—shifting from open retrieval to graph-anchored candidate verification—rather than from the graph directly producing answers.

Our main contributions are:

- A **hierarchical spatio-temporal scene graph index** for long videos, comprising frame-level, segment-level, and global event-level graphs linked by a shared entity registry, together with a bottom-up construction pipeline that builds all three levels offline from VLM-extracted keyframe graphs.
- A **graph-driven reasoning agent** that follows a propose–verify–answer paradigm within an observe–reason–act loop, with graph-guided candidate proposal, frame-level visual verification, caption-based fallback, and uncertainty-aware control strategies that convert open-ended search into structured candidate verification.
- **Comprehensive evaluation** on four long-video benchmarks (LVBench, LongVideoBench, Video-MME, and EgoSchema), with strictly matched ablation and behavioral analyses on LVBench confirming that the gains stem from restructuring the search paradigm rather than the graph directly producing answers.

2 Related Work

Long-video understanding. Existing methods fall into two broad classes. The first extends context capacity or compresses visual redundancy through longer context windows, token compression, or hierarchical video representations [**? ? ? ?**]; these improve input coverage but still must locate sparse, question-relevant evidence within compressed representations. The second treats long-video understanding as a retrieval or search problem, progressively mining relevant information through selective sampling, external databases, or agentic search [**? ? ?**]. SGDAgent belongs to the second class but differs in focus: rather than performing stronger query reformulation over the caption database alone, we ask whether the *evidence space itself* can be restructured—extending the caption database with a structured scene graph index so that the main retrieval steps become candidate proposal processes grounded in entities, relations, and temporal constraints.

Agentic video reasoning. A growing body of work models long-video understanding as an agentic reasoning problem. VideoAgent [**?**] controls multi-round frame retrieval and self-reflection through an LLM-driven state–action–observation loop. DVD [**?**] establishes agentic search over a multi-granularity video database, letting the LLM adaptively invoke tools at different temporal scales. VideoSeek [**?**] designs a multi-granularity toolkit that mirrors human seeking behavior. These methods progressively improve the flexibility and efficiency of the agent's search strategy, yet they all operate over the same evidence space—flat captions or raw frames. SGDAgent extends the agent's action space from caption retrieval to a cascaded process of graph-guided candidate proposal, visual verification, and controlled fallback, thereby transforming the core search space from an open-ended caption space to a graph-anchored candidate space.

Temporal grounding and verification. VideoMind [**?**] argues that effective video reasoning requires a phased process of moment localization, boundary verification, and answer generation, implementing this through four specialized roles (Planner, Grounder, Verifier, Answerer) on a unified backbone. SGDAgent shares the view that long-form video QA benefits from a phased *localize–verify–answer* process, but differs in mechanism: rather than training a dedicated grounding backbone, we use multi-level scene graphs to progressively propose candidates at event, segment, and frame granularities, then delegate final confirmation to visual inspection. Grounding is thus jointly accomplished by the structured graph index and online agent strategy rather than solely by a learned module.

3 Methodology

SGDAgent augments the caption database with a *hierarchical spatio-temporal scene graph index* that converts open-ended caption search into graph-anchored candidate verification. As illustrated in Figure **??**, the system consists of three components. **(i) Hierarchical scene graph representation** (Section **??**): a three-level structured index comprising frame-level graphs G^f, segment-level graphs G^s, and a global event graph G^g, which together with the caption database $C(V)$ form an augmented video database $D(V) = \{C(V),\ G^f,\ G^s,\ G^g\}$. **(ii) Offline graph construction** (Section **??**): a bottom-up pipeline that extracts frame-level entity–relation graphs via a VLM, merges them into segment-level temporal graphs through sliding-window aggregation, and further clusters segments into a global event graph with explicit event edges. **(iii) Graph-driven agent** (Section **??**): an online reasoning agent that operates in an observe–reason–act loop over the augmented database, following a *propose–verify–answer* paradigm—graph queries propose spatiotemporally grounded candidate windows, frame-level visual inspection verifies candidates on raw frames, and an uncertainty-aware controller governs evidence sufficiency and fallback. Algorithm **??** summarizes the online search procedure.

3.1 Hierarchical Scene Graph Representation

SGDAgent organizes video-level knowledge as the caption database augmented with a three-level spatio-temporal scene graph index (Figure **??**):

$$D(V) = \{C(V),\ G^f,\ G^s,\ G^g\}, \tag{1}$$

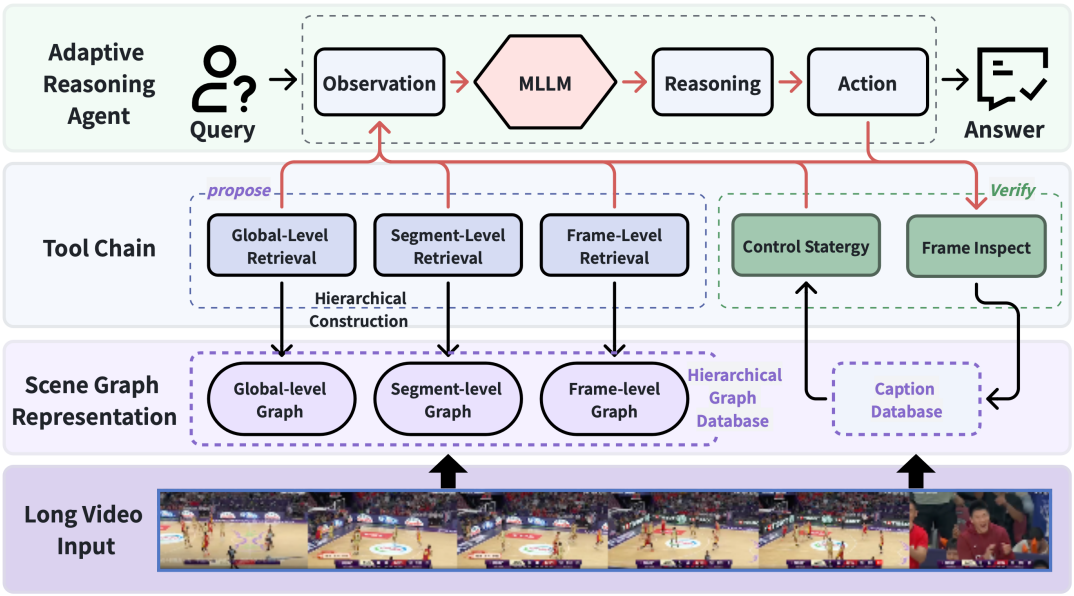


Figure 1: Overview of SGDAgent. Long videos are converted into a scene-graph-augmented video database with global-, segment-, and frame-level graphs plus a caption database. An adaptive reasoning agent then searches this database through a propose–verify tool chain within an observe–reason–act loop.

Algorithm 1 Graph-Guided Search

Require: Augmented database $D(V) = \{C(V), G^f, G^s, G^g\}$, question q
Ensure: Answer $\hat{a}$

```
Parse q into structural constraints φ(q) and routing type π(q)
if q involves long-range temporal or event-level cues then
    Propose event candidates on G^g; cascade to supporting segments on G^s
else
    Propose candidate windows directly on G^s
end if
repeat
    if fine-grained object anchoring is needed then
        Refine candidates on G^f
    end if
    Z_q ← verify top candidates via frame_inspect on raw frames
    if gating conditions satisfied then
        return Answer(q, Z_q)
    else if graph evidence weak or stagnation detected then
        Fall back to caption search or global browse
    end if
until answer committed or budget exhausted
Force terminal convergence from accumulated evidence
```

where $C(V)$ is the caption database inherited from DVD for open semantic retrieval; $G^f = \{G_i^f\}_{i=1}^N$ is the frame-level graph sequence; $G^s = \{G_j^s\}_{j=1}^M$ is the segment-level graph sequence; and $G^g = (V^g, \mathcal{E}^g)$ is the global event graph, with event node set V^g and event edge set $\mathcal{E}^g$. All three graph levels share a video-level entity registry E^{video}, where each entry stores a unique ID, entity type, attributes, description, and temporal span (first and last appearance), serving as the cross-scale identity anchor.

Frame-level graph. Each frame-level graph captures the entity–relation snapshot at a single keyframe:

$$G_i^f = (E_i^f,\ R_i^f,\ t_i,\ c_i), \tag{2}$$

where $E_i^f \subseteq E^{\text{video}}$ is the set of entities visible in the frame, R_i^f is the set of relation triples (s, p, o) with $s, o \in E_i^f$ and predicate p, t_i is the timestamp, and c_i is the aligned caption context. Frame-level graphs provide the finest spatial granularity, anchoring individual objects and their local relations to specific time points.

Segment-level graph. Each segment-level graph aggregates entity interactions over a short temporal window:

$$G_j^s = (E_j^s,\ R_j^s,\ [t_j^s, t_j^e],\ F_j,\ s_j), \tag{3}$$

where E_j^s is the active entity set within the segment, R_j^s is the set of temporal relations, $[t_j^s, t_j^e]$ is the time range, F_j is the set of supporting frame IDs, and s_j is a concise segment summary. Each temporal relation is represented as $(s, p, o, t_{\text{start}}, t_{\text{end}}, \mathcal{F})$, binding participants to their temporal extent and supporting frames.

Global event graph. The global event graph $G^g = (V^g, \mathcal{E}^g)$ organizes higher-level event semantics on top of segments. Each event node $n_k \in V^g$ is defined as

$$n_k = (P_k,\ R_k,\ [t_k^s, t_k^e],\ S_k,\ F_k,\ \sigma_k), \tag{4}$$

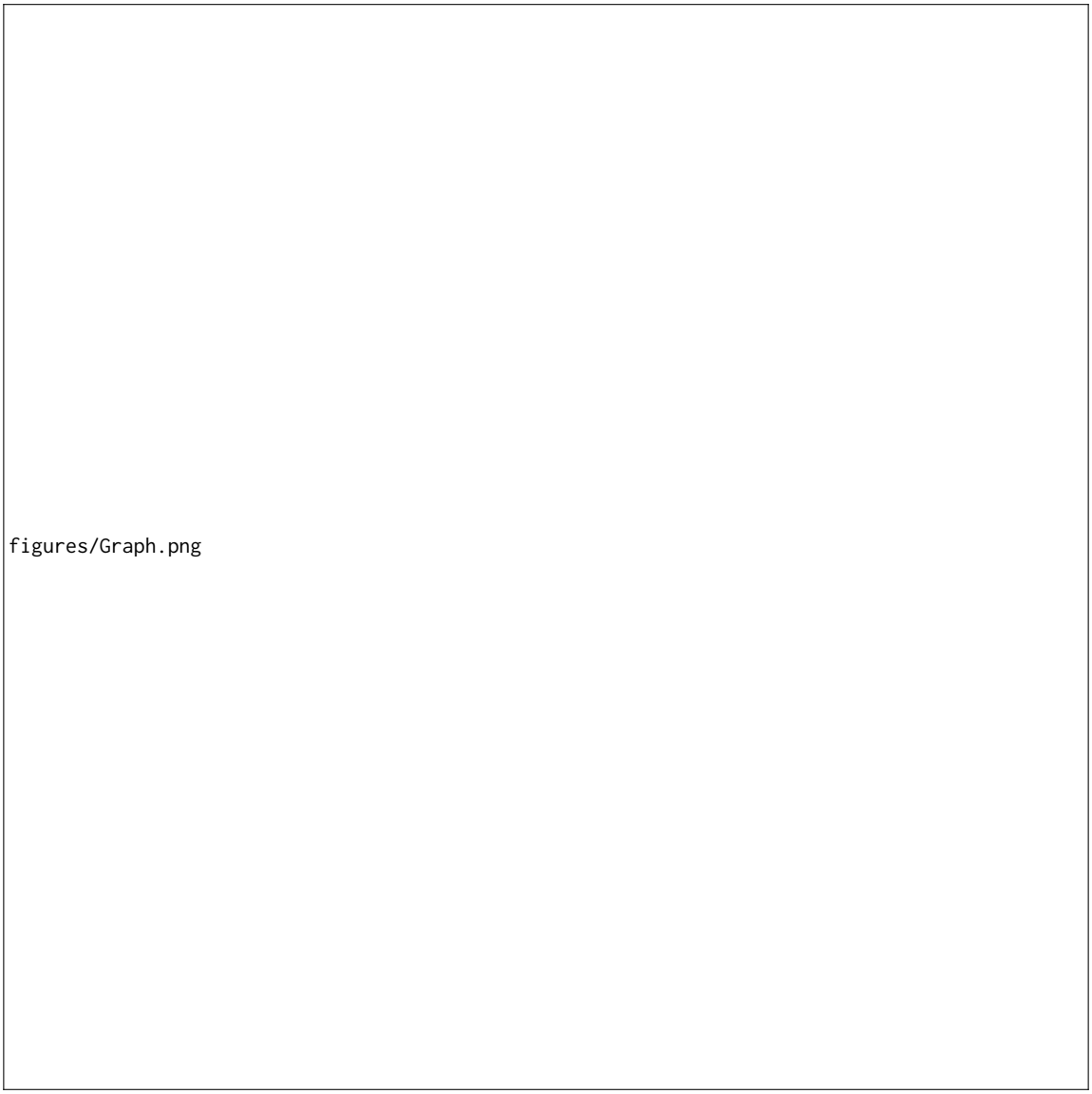


Figure 2: Hierarchical scene graph representation and bottom-up construction pipeline. (a) Keyframes are sampled from the long video. (b) A VLM generates frame-level scene graphs; entity alignment maintains identity consistency via a shared video-level entity registry. (c) The three-level representation: frame-level graphs capture per-keyframe entity–relation snapshots, segment-level graphs aggregate temporal interactions within sliding windows, and the global event graph organizes high-level event semantics with explicit event edges.

where P_k is the participant entity set, R_k is the salient relation set, $[t_k^s, t_k^e]$ is the event time range, S_k is the set of supporting segment IDs, F_k is the supporting frame set, and σ_k is the event summary. Event nodes are typed as either *unit events* (compact local episodes) or *group events* (longer episodes composed of multiple units). The event edge set $\mathcal{E}^g$ contains typed edges—`before`, `overlaps`, `entity-carryover`, `same-thread`, and `parent-child`—encoding cross-segment event chains and hierarchical event structure.

3.2 Hierarchical Scene Graph Construction

The offline phase builds the three graph levels bottom-up—frame, then segment, then global event—so that online reasoning can perform cascaded retrieval over a pre-built structured index.

Frame graph extraction and entity alignment. Keyframes are uniformly sampled at a fixed interval. For each keyframe, a VLM produces a structured scene graph in a single pass, outputting typed entities with attributes and relation triples. Entity identity across frames is maintained by greedily aligning each new entity to the video-level registry E^{video} via a weighted combination of

name similarity, description similarity, and type consistency; entities whose score exceeds a threshold τ are merged, and the rest are registered as new. When the VLM fails to extract any entity, coarse-grained entities are recovered from temporally neighboring captions to prevent empty-graph gaps.

Segment graph merging. Frame-level graphs are aggregated into segment-level graphs using overlapping sliding windows. Within each window, frame-level relation triples sharing the same (s, p, o) signature are merged into a single temporal relation whose extent spans the earliest to the latest supporting frame; supporting frame IDs and occurrence counts are preserved. Each segment graph is equipped with a concise textual summary, and overlap links between adjacent windows support cross-boundary retrieval.

Event graph construction. Temporally proximate and semantically coherent segments are clustered into events. Merging decisions are governed by temporal gap, shared entity count, relation overlap, and predicate overlap; overly long or discontinuous clusters are split into finer-grained unit events. Group events are formed from multiple units with `parent-child` edges. Pairwise event edges—`before`, `overlaps`, `entity-carryover`, and `same-thread`—are inferred from temporal ordering and entity sharing, completing the event-level retrieval graph.

Quality control. Throughout construction, lightweight constraints are applied: entity blacklist filtering removes generic or uninformative entities, predicate whitelist constraints suppress noisy relations, and schema validation enforces structural consistency of each graph level. At the event level, confidence and uncertainty estimates are attached to each node to inform downstream gating decisions.

3.3 Hierarchical Scene Graph Driven Agent

SGDAgent retains DVD's observe–reason–act backbone and its caption-based tools (`clip_search`, `global_browse`, `frame_inspect`) as fallback channels, while introducing graph query tools at each level (`global_graph_query`, `segment_graph_query`, `frame_graph_query`) as the primary candidate proposal interface. The full tool set is $\mathcal{T} = \{GG, SG, FG, FI, \textsc{Answer}\}$, where GG, SG, FG query the three graph levels and FI performs visual verification on raw frames.

Reasoning loop. At each reasoning step t, the agent observes the current state—the question, accumulated tool returns, reasoning history, and control signals—and the backbone LLM selects the next action: querying a graph level to propose candidates, invoking `frame_inspect` to verify a candidate window, falling back to caption search, or committing an answer. Tool outputs are appended to the state for the next step, forming a closed observe–reason–act loop (Algorithm **??**). Unlike pipeline methods with a fixed workflow, the search path is driven jointly by the question type, graph retrieval results, and runtime control signals, allowing the agent to adaptively enter the graph hierarchy at any level and drill down as needed.

Proposal. The system first extracts lightweight structural constraints from the question—entity mentions, relation or action cues, temporal anchors, and whether fine-grained visual attributes are involved—using rule-based parsing. These constraints drive a top-down cascade: for long-range temporal or event-level questions, GG retrieves matching events and cascades their supporting segments to SG; for local relation or short-range questions, candidates are proposed directly from SG; and when object-level anchoring is needed, FG further narrows candidates to specific frames. Candidates at each level are ranked by a composite score

$$S(w \mid q) = S_{\text{lex}}(w, q) + S_{\text{soft}}(w, q) + S_{\text{temp}}(w, q), \tag{5}$$

where S_{lex} measures lexical match over entities, predicates, and summaries, S_{soft} captures embedding-based semantic similarity, and S_{temp} scores temporal consistency with explicit time references or natural-language temporal anchors in q.

Verification. Graph-proposed candidates specify *where to look* but do not constitute terminal evidence. For each top candidate window, the agent invokes `frame_inspect` to sample raw frames and perform visual question answering conditioned on the query. This two-stage separation transforms frame inspection from a blind searcher over the entire video into a focused verifier over graph-localized regions.

Uncertainty-aware control. Three lightweight mechanisms prevent premature commitment and indefinite looping. (1) *Query-type routing*: the question is classified into routing types (summary, temporal chain, scene localization, fine detail, other) that determine the initial graph level and the minimum evidence requirements—e.g., fine-detail questions require narrow-window visual inspection before an answer can be committed. (2) *Confidence gating*: when graph queries return weak or stagnating signals, the system falls back to caption-based tools before resuming graph queries. (3) *Terminal convergence*: when the budget is exhausted, a forced procedure extracts the best-supported option from the accumulated trace, preventing well-evidenced trajectories from being lost as FTT outcomes. Section **??** separately evaluates routing and gating versus terminal convergence.

4 Experiments

4.1 Experimental Setup

Dataset. Our primary benchmark is **LVBench** [**?**], which contains 1,549 four-choice video QA questions over long-form videos. The system receives a video, a natural-language question, and four candidate answers, and must return exactly one option; failure to do so is recorded as a failed-to-terminate (FTT) outcome.

Implementation details. The reasoning backbone is `o4-mini` (temperature 0) with a maximum of 15 reasoning steps per question. Frame-level scene graphs (Stage 1) are extracted by prompting `gpt-4.1` on keyframes sampled at one frame every 5 seconds (up to 120 per video). Segment-level graphs (Stage 2) use 30-second sliding windows with 5-second overlap. Local visual verification is performed by `gpt-4.1-mini`, with a budget of 50 frames per `frame_inspect` call. Graph retrieval returns at most 5 event-level, 6 segment-level, and 8 frame-level candidates. To ensure fair matched-setting comparison, all internally compared methods share the same reasoning backbone, frame-inspect VLM, caption source, video database, candidate answer set, and budget limits; all methods are permitted to produce FTT outcomes.

Evaluation metrics. We report four complementary metrics: *end-to-end accuracy* (e2e_acc), the fraction of all questions answered correctly; *answered accuracy* (ans_acc), accuracy computed only over questions for which the system produced a valid option; *answer rate*, the fraction of questions that received a valid option; and *FTT rate*, the fraction of questions where the system failed to commit an answer due to empty output, runtime failure, or trajectory non-convergence.

4.2 Main Results

Performance positioning. Table **??** compares SGDAgent with published results across four long video understanding benchmarks. Since these methods differ in backbone, caption source, and evaluation protocol, the numbers should be read as reference positions rather than controlled comparisons; the strict conclusions of this paper rest on the matched-setting ablations in Section **??**.

Three observations are noteworthy. First, *the structured graph index narrows the gap between a weaker and a stronger reasoning backbone.* SGDAgent with `o4-mini` achieves 73.0% on LVBench and 71.2% on LongVideoBench, nearly matching DVD with the substantially stronger `o3` backbone (74.2% and 71.6%, respectively). This suggests that when the evidence space is restructured from flat captions into entity-, relation-, and temporally-grounded candidates, the agent needs less raw reasoning power to locate the right evidence—the graph compensates for what the weaker backbone lacks in open-ended search capability. Second, when paired with the same `o3` backbone, SGDAgent consistently outperforms DVD on the three benchmarks that emphasize long-duration and multi-entity reasoning (LVBench 75.9 vs. 74.2, LongVideoBench 74.0 vs. 71.6, Video-MME Long 69.2 vs. 68.6), confirming that the gains from graph-guided candidate verification are additive to backbone strength rather than merely compensatory. Third, on EgoSchema—where videos are only 3 minutes and the graph hierarchy collapses to a handful of segments—SGDAgent (77.4) slightly trails DVD (78.0), consistent with the expectation that the structured proposal layer contributes most when the search space is large and evidence is temporally sparse.

Matched-setting analysis. To isolate the contribution of each component under controlled conditions, Table **??** reports matched-setting results on LVBench where all methods share the same backbone, caption source, and budget.

Effect of graph-augmented database. Combining the caption database with the scene graph index (SGDAgent-base) raises end-to-end accuracy by nearly 10 points over the caption-only baseline, while the full system with control strategies reaches 73.0%. Notably, using scene graphs alone (*SG only*) is weaker than the combined configuration: its end-to-end accuracy exceeds caption-only by only 3 points and its answered accuracy is actually lower. This confirms that the gain comes not from replacing captions with graphs, but from the complementary combination—the caption database preserves open semantic coverage while graphs contract the search space into verifiable candidate regions. The following ablation study further dissects which graph levels and control strategies drive these gains.

4.3 Ablation Study

Having established SGDAgent's overall effectiveness in Section **??**, we now ablate its two core design dimensions under strictly matched settings: graph level depth (Sections **??**–**??**) and agent control strategies (Section **??**).

Effect of graph level depth. Table **??** dissects the contribution of each graph level. The largest accuracy gain comes from adding the segment-level graph (Stage 1 → Stage 1+2), which provides the structured candidate discovery chain of "frame-level localization + segment-level temporal organization." Adding the global event graph (Stage 3) does not further improve accuracy, but it halves the average number of LLM calls and reduces frame-inspection calls by 5×: event-level proposal narrows the search to relevant segments in one step, eliminating the need for exhaustive segment-by-segment exploration. Stage 3 therefore contributes primarily as an efficiency layer rather than an accuracy layer.

Effect of agent control strategies. Table **??** evaluates the two control strategies introduced in Section **??**. Strategy R (routing and gating) improves answered accuracy from 72.2% to 78.0% while moderately reducing the FTT rate, indicating that it primarily raises answer quality on already-answerable questions by ensuring fine-grained questions receive targeted visual verification rather than relying on graph hits alone. Strategy T (terminal convergence) addresses a different axis: it substantially compresses the FTT rate by recovering valid answers from trajectories that gathered sufficient evidence but failed to produce a well-formatted response. The combination of both strategies yields the best balance: end-to-end accuracy reaches 73.0% with an FTT rate of only 1.6%, confirming that routing and gating improve *how well* the agent answers, while terminal convergence improves *whether* it answers.

Efficiency analysis. Beyond accuracy, a central claim of this paper is that graph-anchored candidate verification restructures the search process itself. Table **??** compares per-question search cost between the caption-only baseline and SGDAgent variants across all questions with complete trajectory traces.

Two patterns emerge. First, introducing segment-level graph retrieval (S1+2) substantially increases tool usage—LLM calls rise from 3.7 to 10.6 and frame inspections from 1.1 to 5.3—indicating that the accuracy gains come from deeper structured candidate discovery and stronger local verification, not from lightweight reranking of existing trajectories. Second, adding the global event graph (full system) cuts these costs roughly in half while maintaining the accuracy gain: LLM calls drop to 3.4 and frame inspections to 1.0. This confirms that the event-level proposal layer narrows search to relevant segments in fewer steps, serving as an effective efficiency mechanism on top of the accuracy-oriented segment-level graph.

4.4 Analysis on Agentic Reasoning Behavior

The ablation results show *what* improves; this section examines *how* the search behavior changes by comparing full reasoning trajectories between the caption-only baseline and SGDAgent.

Trajectory pattern shift. Figure **??** contrasts representative trajectories for three question types. In the caption-only baseline, the agent typically begins with a broad global browse or multiple caption searches, hoping to encounter relevant descriptions; when initial hits are vague, it rewrites queries and repeats similar

Table 1: Comparison on long video understanding benchmarks. LVBench and Video-MME (Long) target hour-scale videos; LongVideoBench covers medium-to-long durations; EgoSchema evaluates egocentric temporal reasoning over 3-minute clips. Methods are grouped by type: commercial VLMs, open-source VLMs, and video agents.

Method	Backbone	LVBench *full*	LongVideoBench *Val, overall*	Video-MME *Long, w/o sub*	EgoSchema *Val 500*
Commercial VLMs					
Gemini-1.5-Pro [?]	Gemini	33.1	64.0	58.6	–
GPT-4o [?]	GPT-4o	48.9	66.7	60.9	70.4
OpenAI o3 [?]	o3	57.1	67.5	60.6	63.2
Gemini-2.5-Pro [?]	Gemini	67.4	–	–	–
Open-Source VLMs					
Qwen2.5-VL-72B [?]	72B	47.7	60.7	–	–
InternVL2.5-78B [?]	78B	43.6	63.6	–	–
Seed1.5-VL-Thinking [?]	Seed-VL	64.6	–	–	–
Video Agents					
VideoAgent [?]	–	29.3	–	–	63.2
MR. Video [?]	Gemini-Flash	60.8	–	61.8	73.0
VideoMind [?]	7B	40.8	–	–	–
VideoSeek [?]	GPT-5	68.4	–	–	–
DVD [?]	o4-mini	70.2	–	–	–
DVD [?]	o3	74.2	71.6	68.6	78.0
Ours					
SGDAgent	o4-mini	**73.0**	71.2	65.1	73.9
SGDAgent	o3	**75.9**	74.0	69.0	77.4

Table 2: Matched-setting results on LVBench. All methods share the same backbone, caption source, and budget.

Method	e2e_acc	ans_acc	ans rate	FTT rate
Caption only	54.8%	65.7%	83.5%	16.5%
SG only	58.0%	69.1%	83.9%	16.1%
SGDAgent-base	64.4%	72.2%	89.2%	10.8%
SGDAgent (full)	73.0%	74.2%	98.4%	1.6%

searches, often exhausting the step budget without converging. SGDAgent replaces this open-ended exploration with a focused propose–verify sequence: a graph query directly returns candidate time windows grounded by entities, relations, and temporal ranges, after which frame inspection verifies visual details within these narrow windows. The resulting trajectories are both shorter (fewer reasoning steps on average) and more targeted (higher proportion of steps that produce qualified new evidence).

Quantitative behavior breakdown. Automated heuristic analysis of trajectory tool-call patterns identifies 168 questions that flip from wrong or FTT to correct (*improved-to-correct*) versus 70 that regress in the opposite direction (*correct-to-wrong*), yielding a net gain of +98 questions. The three most frequent improvement patterns are: (1) the agent discovers a relevant segment through a graph query that the baseline never reached; (2) the agent follows a graph-then-visual-confirm sequence that replaces a broad caption search with a focused verification; and (3) the agent performs additional visual confirmations on graph-localized regions, reaching higher-confidence evidence. These patterns confirm that accuracy gains are driven by the propose–verify workflow restructuring the evidence gathering process, not by the graph directly producing answers.

Case study. We highlight three trajectories that illustrate specific strengths and limitations of graph-driven reasoning.

Temporal-constrained event QA. A question asks what happened after two characters first came ashore, with an explicit time reference. The caption-only agent performed only coarse global browsing and failed to converge. SGDAgent's segment graph directly returned a high-scoring candidate within the target time window; a single focused frame inspection confirmed the answer in five steps (Figure **??**, left). This case demonstrates how graph-anchored proposal compresses search from global caption space to a few temporally consistent regions.

Fine-grained attribute recognition. A question asks the color of a car's steering wheel. The baseline performed one broad frame inspection and answered "black" (wrong). SGDAgent first used the frame-level graph to anchor "steering wheel" to a specific keyframe; the control layer then blocked premature termination, requiring repeated narrow-window inspections until observations converged to "white cover" (correct; Figure **??**, center). This case shows the interplay between graph localization and the routing and gating strategy.

Failure case: correct localization, wrong judgment. A question asks which body part a character first kissed another on. SGDAgent quickly localized the relevant segment and committed an answer, but frame inspection judged "cheek" while the ground truth is "forehead" (Figure **??**, right). This illustrates SGDAgent's capability boundary: the graph module reliably addresses *where to look* and

Table 3: Ablation study results on LVBench. All methods share the same backbone, caption source, and budget.

Method	e2e_acc	ans_acc	FTT rate	LLM calls	FI calls
Caption-only	54.8%	65.7%	83.5%	4.92	2.38
SG-only	58.0%	69.1%	83.9%	4.92	2.38
Caption-only					
+ Graph Stage 1	59.2%	69.0%	83.5%	4.92	2.38
+ Graph Stage 1 + 2	64.6%	72.0%	83.5%	10.6	5.34
+ Graph Stage 1 + 2 + 3 (= SGDAgent-base)	64.4%	72.2%	83.5%	5.34	1.01
SGDAgent-base					
+ Strategy R	71.5%	78.0%	91.7%	4.92	2.38
+ Strategy R + T	71.2%	72.1%	98.7%	4.92	2.38
+ Strategy R + G	70.2%	75.6%	91.0%	4.92	2.38
+ Strategy R + G + T (= SGDAgent-full)	73.0%	74.2%	98.4%	4.92	2.38

Table 4: Ablation on graph level depth. "LLM calls" and "FI calls" are per-question averages on LVBench with complete trajectory traces.

Method	e2e_acc	ans_acc	LLM calls	FI calls
Stage 1 only	59.2%	69.0%	4.92	2.38
Stage 1 + 2	64.6%	72.0%	10.6	5.34
Stage 1 + 2 + 3	64.4%	72.2%	5.34	1.01

Table 5: Effect of control strategies. R: query-type-aware routing and terminal evidence gating. T: terminal convergence. G: confidence gating. SGDAgent-base uses the graph-augmented database with all three graph levels but no control strategies.

Method	e2e_acc	ans_acc	ans rate	FTT rate
SGDAgent-base	64.4%	72.2%	89.2%	10.8%
+ Strategy R	71.5%	78.0%	91.7%	8.30%
+ Strategy R + T	71.2%	72.1%	98.7%	1.30%
+ Strategy R + G	70.2%	75.6%	91.0%	9.00%
+ Strategy R + G + T (= SGDAgent full)	73.0%	74.2%	98.4%	1.60%

Table 6: Search efficiency comparison (per-question averages).

Method	ans_acc	LLM calls	tool calls	FI calls
Caption only	65.7%	3.69	3.53	1.10
SGDAgent S1+2	72.0%	10.63	10.34	5.34
SGDAgent (full)	74.2%	3.44	4.21	1.01

when to stop searching, but cannot correct errors in the local visual judgment itself.

5 Conclusion

This paper proposes SGDAgent for augmenting the caption database with a hierarchical scene graph index in long-form video question answering. The core conclusion is not that "the graph module brings some additional improvement" but a stronger search paradigm thesis: **SGDAgent converts open-ended caption search into graph-anchored candidate verification.**

This thesis uniformly explains the main method design and experimental results. First, it explains why the graph-augmented database is necessary: caption alone retains an overly large open search space, while graph alone is insufficient to cover open semantic descriptions, so both must complement each other. Second, it explains why SGDAgent can significantly reduce FTT: when search transforms from open-ended recall to candidate verification, the agent more easily forms stable trajectories and commits terminal judgments. Finally, it explains why accuracy improves: SGDAgent not only answers more questions but also achieves higher verification quality on already-answered samples.

From current results, SGDAgent's primary gains come from Stage 1 and Stage 2's local structured candidate discovery, while Stage 3 further supplements event-level candidate proposal, event chain modeling, and event-to-segment cascade. Future work can proceed along two directions: improving local visual verification reliability to reduce residual errors from "FTT to wrong answer" conversions, and enhancing the global event graph's event abstraction and chain retrieval capabilities for more stable long-range event reasoning.

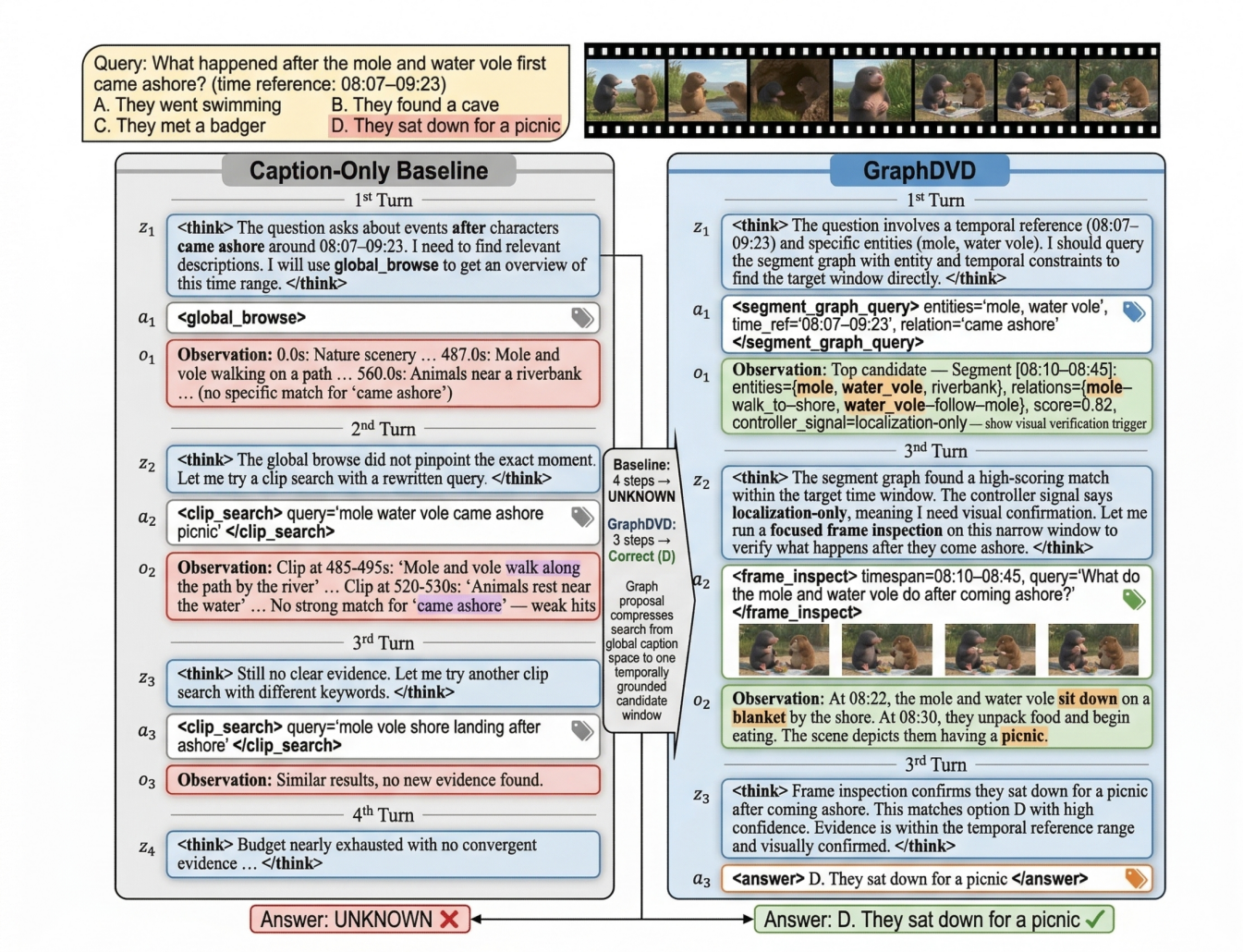


Figure 3: Trajectory comparison between the caption-only baseline and SGDAgent on three representative question types. Graph query tools (blue) replace broad caption searches (gray) and guide focused frame inspections (green) to narrow candidate windows, yielding shorter and more targeted trajectories. The rightmost case illustrates a failure mode where correct localization does not prevent an incorrect visual judgment.